Lecture 21

Gauge fields: connections, curvature and holonomy

Objectives. *A bundle has local frames, but differentiating their coefficients depends on the frame. A connection corrects this derivative and specifies parallel transport. We derive its change law and show how curvature measures transport around a small loop. A flat connection can still have nontrivial transport around a hole. The Aharonov–Bohm phase is this effect for a charged particle.*

By the end of this lecture you will be able to:

- Derive the connection and curvature laws under a change of frame.
- Compute parallel transport with the correct multiplication order.
- Relate a small counter-clockwise loop to its curvature.
- Compute the flat Aharonov–Bohm holonomy and its gauge-invariant phase.

Unit 21A Connection and curvature

21.1 Covariant derivatives

Let $E \to M$ be a smooth rank-r complex vector bundle. Write $\Gamma(E)$ for its smooth sections and $\Omega^k(M;E)$ for its E-valued k-forms. In a frame, the latter are columns of ordinary k-forms. Recall Lecture 20's row frame e and coefficient column σ, with $s = e\sigma$. A change of frame $e' = eh$ gives $\sigma' = h^{-1}\sigma$. The ordinary derivative therefore obeys

$$\mathrm{d}\sigma' = h^{-1}\mathrm{d}\sigma + \mathrm{d}(h^{-1})\sigma.$$

The extra term means that the derivative of a coefficient column does not transform as an E-valued one-form. Even a constant column can become nonconstant after a frame change. A connection supplies a derivative independent of this choice. For a charged wave amplitude, it supplies a rule for comparing phases in different fibres.

Definition 21.1. A *connection* is a complex-linear map

$$\nabla\colon \Gamma(E) \longrightarrow \Omega^1(M;E), \qquad \nabla(fs) = \mathrm{d}f \otimes s + f\nabla s \quad \text{for } f \in C^\infty(M,\mathbb{C}).$$

Here $(\mathrm{d}f \otimes s)_p(X) = \mathrm{d}f_p(X)s(p)$ for $p \in M$ and $X \in T_pM$. Thus ∇s assigns a fibre vector to each tangent direction. We write $(\nabla s)(X) = \nabla_X s$, as for tangent fields in Lecture 16.

Locality. We construct a *bump function*, a smooth function equal to one near a chosen point $p \in M$, with support inside a chosen neighbourhood U. Take a coordinate ball centred at p with $\overline{B_R} \subset U$ and $0 < r < R$. Define $\rho(t) = e^{-1/t}$ for $t > 0$ and $\rho(t) = 0$ for $t \le 0$. Every positive-side derivative is $e^{-1/t}$ times a polynomial in $1/t$, by induction. All tend to zero at zero, so ρ is smooth. In the ball's coordinates the desired function is

$$f(x) = \frac{\rho(R^2 - |x|^2)}{\rho(R^2 - |x|^2) + \rho(|x|^2 - r^2)}.$$

The denominator is positive. The function is one on B_r and zero outside B_R. Its support is compact inside the chart, so extension by zero is smooth on M.

The connection is local. If s vanishes near p, choose such an f supported where $s = 0$. The Leibniz rule applied to $fs = 0$ gives $(\nabla s)_p = 0$. A section defined on U can instead be multiplied by a bump supported in U, equal to one near p, and extended by zero. Locality makes its derivative at p independent of the extension. This defines the local operator with the same Leibniz rule and commutes with restriction. In a local frame, define the matrix of one-forms A by $\nabla e = eA$. The Leibniz rule gives

$$\nabla s = eD\sigma, \qquad D\sigma = \mathrm{d}\sigma + A\sigma.$$

Definition 21.2. The matrix A in $\nabla e = eA$ is the local *gauge potential*.

It is a matrix of one-forms, rather than a single global matrix on a nontrivial bundle. Lecture 16's tangent connection is the real version of this definition, with real fibres, real linearity and real-valued functions. The figure shows one fixed vector in three rotating frames in this real rank-two case.

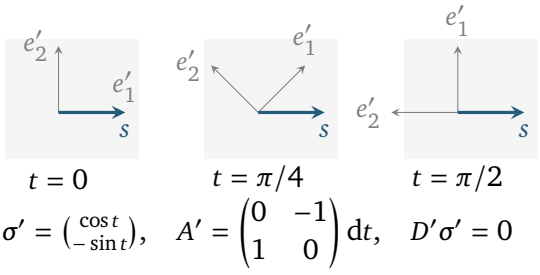


Figure 21.1: A fixed vector has changing coefficients in rotating frames. Its covariant derivative remains zero because the connection term cancels the ordinary derivative of its coefficients.

21.2 Frame changes and gauge transformations

Let $h\colon U \to \mathrm{GL}(r, \mathbb{C})$ be a smooth change of frame, $e' = eh$. The symbol h denotes a frame change. Planck's constant is h_{P}. The same section has coefficients $\sigma' = h^{-1}\sigma$. Differentiating the new frame gives

$$\nabla(eh) = e(Ah + \mathrm{d}h) = ehA', \qquad A' = h^{-1}Ah + h^{-1}\mathrm{d}h. \tag{21.1}$$

The extra derivative compensates for the changing frame. The following diagram expresses the resulting covariance.

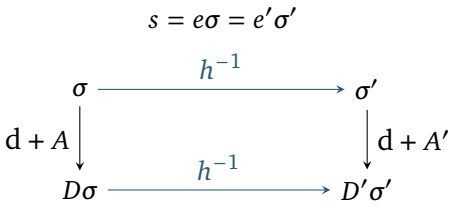


Figure 21.2: Both a section's coefficient column and its covariant derivative change by h^{-1}. The connection law makes the square commute.

Indeed, $\mathrm{d}(h^{-1}) = -h^{-1}(\mathrm{d}h)h^{-1}$ implies $D'\sigma' = h^{-1}D\sigma$ after the two derivative terms cancel. Conversely, local matrices obeying this change law on overlaps define a connection, because these local derivatives agree as bundle-valued one-forms. The compatibility law is Nakahara's (10.9).[1]

Definition 21.3. For an atlas with transitions in a matrix group G, its *G-frames* are the right translates $e_i a$, with $a \in G$. Require A to take values in its Lie algebra in these frames.

We use these frames for group-valued transport in §21.4.

The G-frames of E form a principal bundle by Lecture 20's cocycle construction.

Definition 21.4. A *gauge transformation* is an equivariant smooth automorphism of this principal bundle over the identity of M.

In local sections s_i, write it as $s_i(x)a \mapsto s_i(x)u_i(x)a$. The right-hand group coordinate is a. Since $s_j = s_i g_{ij}$, compatibility is precisely

$$u_j = g_{ij}^{-1} u_i g_{ij}.$$

A chosen global trivialization identifies these transformations with maps $M \to G$. For an abelian group the compatibility law reduces to $u_i = u_j$, even on a nontrivial bundle. For a nonabelian group, arbitrary global matrix maps need not satisfy it. A passive frame change describes the same connection. On E, the corresponding automorphism acts on columns by $\sigma_i \mapsto u_i \sigma_i$. Transporting the connection so that $D_i'(u_i\sigma_i) = u_i D_i \sigma_i$ gives

$$A_i' = u_i A_i u_i^{-1} + u_i \mathrm{d}(u_i^{-1}).$$

This follows by differentiating $u_i\sigma_i$ and cancelling $\mathrm{d}u_i$. It is the frame law with $h = u_i^{-1}$, the active description.

The passive and active descriptions keep different data fixed:

	Fixed data	Transformed data
Passive	section and connection	frame and coefficients
Active	frame	section and connection

A passive change can be chosen on one chart. An active transformation must satisfy the overlap compatibility law on the whole bundle. The equality of their local formulas does not remove this global requirement.

[1] We use row frames, so coefficient columns transform by h^{-1}.

Unitary connections.

Definition 21.5. A *Hermitian metric* is a smoothly varying positive Hermitian inner product on each fibre, antilinear in its first argument.

Definition 21.6. A connection is *unitary* if it differentiates this inner product by the product rule.

Definition 21.7. A *unitary frame* is an orthonormal basis in every fibre of its domain.

In such a frame that rule is equivalent to $A^\dagger = -A$. The covariant product rule differs from the ordinary derivative of $\sigma^\dagger\tau$ by the term $\sigma^\dagger(A^\dagger + A)\tau$. Here σ and τ are coefficient columns of two local sections. For a line bundle, A is therefore imaginary.

A charged particle. For charge $q \in \mathbb{R}$, the physical potential of Lecture 15 is the real one-form $A_{\mathrm{em}} = -\phi\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}\mathbf{x}$. The connection convention and its component derivatives are

$$A_{\mathrm{conn}} = -\frac{\mathrm{i}q}{\hbar}A_{\mathrm{em}}, \qquad D_t = \partial_t + \frac{\mathrm{i}q}{\hbar}\phi, \qquad D_i = \partial_i - \frac{\mathrm{i}q}{\hbar}A_i.$$

Here $\hbar > 0$ remains explicit. Thus $-\mathrm{i}\hbar D_i = -\mathrm{i}\hbar\partial_i - qA_i$.

Definition 21.8. Replacing ordinary derivatives by these derivatives is *minimal coupling*.

The real electromagnetic potential and the imaginary connection potential are distinct.

Physical input. For a spinless nonrelativistic particle of mass $m > 0$ on a flat Cartesian patch, the local wave amplitude $\psi(t, \mathbf{x})$ is a coefficient of the charged section. Its squared modulus is the probability density. The charged Schrödinger equation is

$$\mathrm{i}\hbar D_t\psi = \frac{1}{2m}\sum_i(-\mathrm{i}\hbar D_i)^2\psi.$$

Each differential operator acts on everything to its right. In the ideal coherent-arm approximation, the electromagnetic action $q\int_\gamma A_{\mathrm{em}}$ multiplies an arm amplitude by $\exp(\mathrm{i}q\int_\gamma A_{\mathrm{em}}/\hbar)$. The remaining kinetic and apparatus phases are included in that arm's amplitude. We hold them fixed when comparing static solenoid fluxes.

Since $\mathrm{i}\hbar D_t = \mathrm{i}\hbar\partial_t - q\phi$, the same equation in ordinary derivatives is

$$\mathrm{i}\hbar\partial_t\psi = \frac{1}{2m}\sum_i(-\mathrm{i}\hbar\partial_i - qA_i)^2\psi + q\phi\psi.$$

The connection laws imply covariance of this equation.

For electromagnetism take a real smooth function χ and $h = e^{-\mathrm{i}q\chi/\hbar}$. The frame law then reads

$$A'_{\mathrm{em}} = A_{\mathrm{em}} + \mathrm{d}\chi, \qquad \psi' = e^{\mathrm{i}q\chi/\hbar}\psi, \qquad D'\psi' = e^{\mathrm{i}q\chi/\hbar}D\psi.$$

Here ψ is the charged section's coefficient, which transforms inversely to its frame. These equations include the charge sign and agree with the minimal-coupling derivatives. Applying the spatial covariance twice shows that each squared operator in the Schrödinger equation transforms by the same phase as ψ. With the original frame held fixed, the same local formulas describe the active map $u = e^{\mathrm{i}q\chi/\hbar}$. A global real function χ defines this automorphism even when no global frame exists.

21.3 Curvature and the Bianchi identity

For matrix-valued forms, wedge multiplication includes matrix multiplication: $(A \wedge B)_{ab} = \sum_c A_{ac} \wedge B_{cb}$. Extend D to coefficient-valued forms by $D\eta = \mathrm{d}\eta + A \wedge \eta$.

Definition 21.9. The *curvature two-form* is

$$F = \mathrm{d}A + A \wedge A, \qquad F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j].$$

Its action on a coefficient column is $D^2\sigma = F\sigma$. The components mean $F = \frac{1}{2} F_{ij} \mathrm{d}x^i \wedge \mathrm{d}x^j$.

Definition 21.10. For an endomorphism-valued k-form B, the *covariant exterior derivative* is

$$DB = \mathrm{d}B + A \wedge B - (-1)^k B \wedge A.$$

This derivative acts on both the output and input of an endomorphism.

Theorem 21.11 (Local connection laws · *proved*)**.** *A connection has local form $D = \mathrm{d} + A$ and the frame law (21.1). Its curvature transforms by $F' = h^{-1} F h$ and satisfies the Bianchi identity*

$$DF = \mathrm{d}F + A \wedge F - F \wedge A = 0.$$

Proof. The first assertions follow from the Leibniz and frame calculations above. Expanding $D^2\sigma$ cancels $-A \wedge \mathrm{d}\sigma$ against $A \wedge \mathrm{d}\sigma$, leaving $F\sigma$. To check curvature covariance explicitly, put $B = h^{-1} A h$ and introduce the *Maurer–Cartan form* $\vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h$. The product rule and the inverse derivative give

$$\mathrm{d}B = h^{-1}(\mathrm{d}A)h - \vartheta_{\mathrm{MC}} \wedge B - B \wedge \vartheta_{\mathrm{MC}}, \qquad \mathrm{d}\vartheta_{\mathrm{MC}} = -\vartheta_{\mathrm{MC}} \wedge \vartheta_{\mathrm{MC}}.$$

Since $A' = B + \vartheta_{\mathrm{MC}}$, substituting these identities into $\mathrm{d}A' + A' \wedge A'$ leaves only $h^{-1}(\mathrm{d}A + A \wedge A)h$. Finally,

$$\mathrm{d}F = \mathrm{d}A \wedge A - A \wedge \mathrm{d}A, \qquad A \wedge F - F \wedge A = A \wedge \mathrm{d}A - \mathrm{d}A \wedge A.$$

Adding gives the Bianchi identity. These are Nakahara's (10.38)–(10.47) in the present matrix notation. ∎

The value of a potential at a point. Within its matrix Lie algebra, the potential's value at a point can be changed arbitrarily. For a prescribed Lie-algebra-valued covector B_p, choose $h(p) = I$ and $\mathrm{d}h_p = B_p - A_p$. The frame law gives $A'_p = B_p$ and $F'_p = F_p$. Such a first derivative is realized locally by exponentiating a Lie-algebra-valued function that vanishes at p and has the prescribed derivative there. Thus A_p alone is not frame invariant. Curvature changes only by conjugation, so its traces are frame invariant. This is the bundle version of the distinction between Christoffel symbols and curvature in Lecture 16 and Lecture 17.

Dictionary of potentials and curvature. For U(1) the scalar one-form obeys $A \wedge A = 0$. The abelian Bianchi identity is $\mathrm{d}F = 0$. The physical and connection quantities are related by

$$F_{\mathrm{conn}} = -\frac{\mathrm{i}q}{\hbar} F_{\mathrm{em}}, \qquad \mathcal{J}_{\mathrm{conn}} = -\frac{\mathrm{i}q}{\hbar} \mathcal{J}.$$

The real source three-form $\mathcal{J}$ is Lecture 15's current. The connection-valued source $\mathcal{J}_{\mathrm{conn}}$ expresses the same source in the units of F_{conn} for the field equation in Lecture 23. For nonzero charge, the abelian Bianchi identity is precisely Lecture 15's homogeneous Maxwell

equation $\mathrm{d}F_{\mathrm{em}} = 0$. For zero charge the connection is zero and does not encode that electromagnetic equation. For a Hermitian matrix potential A_{phys} with $D = \mathrm{d} - \mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$, the corresponding formulas are

$$A_{\mathrm{conn}} = -\mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}, \qquad F_{\mathrm{conn}} = -\mathrm{i}g_{\mathrm{YM}}(\mathrm{d}A_{\mathrm{phys}} - \mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}} \wedge A_{\mathrm{phys}}).$$

The coupling g_{YM} is a scalar, never a gauge map. The commutator term remains in the nonabelian theory.

For the tangent bundle, the matrix A is Lecture 17's matrix of connection forms ω. The matrix F is its curvature matrix Ω. In particular,

$$A = (\omega^i{}_j), \qquad F = (\Omega^i{}_j), \qquad R(X,Y)(ev) = eF(X,Y)v.$$

Along a curve, the matrix $a(t) = A(\dot\gamma(t))$ is Lecture 16's $C(t)$. Lecture 16's frame-change matrix, there called A, is our h. The identity $DF = 0$ is Lecture 17's second Bianchi identity in this notation. On an oriented surface, write μ_g for the metric area form. This is the form written $\mathrm{d}A$ in Lecture 18. It is distinct from the exterior derivative $\mathrm{d}A$ of the connection potential here. The surface formulas of Lectures 17 and 18 become

$$A = -\omega^1{}_2 J, \qquad F = -K\mu_g J, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Here $\omega^1{}_2$ is the scalar connection form and J is the positive quarter-turn. Lecture 18's rotation angle around a framed disk is $\Theta_D \equiv \int_D K\mu_g \pmod{2\pi}$.

Self-check. State in one sentence what a connection is and why an ordinary derivative of coefficient columns does not replace it.

Answer. A connection is a complex-linear Leibniz derivative of sections into bundle-valued one-forms, introduced to compensate for the term $\mathrm{d}(h^{-1})\sigma$ in the ordinary derivative after a frame change. That term prevents $\mathrm{d}\sigma$ from transforming as a bundle-valued one-form.

Unit 21B Transport, holonomy and the Aharonov–Bohm phase

21.4 Parallel transport and a small square

Definition 21.12. A section along a piecewise smooth path $\gamma\colon [0,1] \to M$ is *parallel* if its covariant derivative along the path vanishes.

In one local frame put $a(t) = A_{\gamma(t)}(\dot\gamma(t))$. Its coefficient equation is $\dot\sigma = -a(t)\sigma$.

Definition 21.13. The *transport matrix* $U(t)$ sends the initial column to this column.

Its initial value is I. Its equation is $\dot U = -a(t)U$. The coefficient equation gives the matrix equation column by column. Linear ODE existence and uniqueness give a solution throughout the compact parameter interval. Unlike Lecture 9's incomplete field $x^2\partial_x$, these linear transport equations have solutions on the whole path interval. The inverse solves $\dot V = Va(t)$, so $\frac{\mathrm{d}}{\mathrm{d}t}(VU) = 0$ and transport is invertible. On successive bundle charts, change coefficients at crossings and compose these solutions. The frame covariance proved above and uniqueness make this independent of subdivisions.

Iterating $U(t) = I - \int_0^t a(s)U(s)\mathrm{d}s$ gives

$$U(t) = I + \sum_{n\geq 1}(-1)^n \int_{0<t_1<\cdots<t_n<t} a(t_n)\cdots a(t_1)\mathrm{d}t_1\cdots\mathrm{d}t_n.$$

The nth term has norm at most $K^n t^n/n!$ when $\|a\| \leq K$.

Definition 21.14. The convergent series above is the *path-ordered exponential*.

Its norm is at most e^{Kt}, so no finite-time blow-up occurs. Later times stand on the left. In a unitary frame for a unitary connection, $\frac{\mathrm{d}}{\mathrm{d}t}(U^\dagger U) = 0$, so transport is unitary. For a matrix Lie group G, the equation $\dot{U}U^{-1} = -a(t) \in \mathfrak{g}$ is a right-invariant ODE on G whose G-valued solution extends over the path interval by uniform short-time existence at the identity and restarting by right translations on finitely many subintervals. Uniqueness of the linear ODE in $\mathrm{GL}(r,\mathbb{C})$ identifies this solution with the transport matrix, so transports and based-loop holonomies belong to G. For scalar A this reduces to $U(\gamma) = \exp(-\int_\gamma A)$ in a frame covering the path. This agrees with the U(1) transport of *Lie Groups*, Lecture 27.[2] The figure shows a parallel phase at successive parameter values.

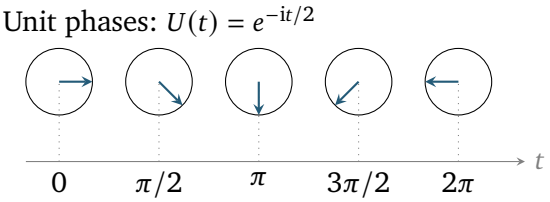


Figure 21.3: The circles show unit phases with $A = \mathrm{i}\mathrm{d}t/2$ and transport $U(t) = e^{-\mathrm{i}t/2}$. The phase turns clockwise and reaches -1 at $t = 2\pi$.

Definition 21.15. The *holonomy* $\mathrm{Hol}(\gamma)$ is the transport around a based loop.

Coefficient covariance $D'\sigma' = h^{-1}D\sigma$ and uniqueness show that changing the base frame conjugates it.

Definition 21.16. The trace of a holonomy is its *Wilson loop*.

Conjugation invariance of the trace makes the Wilson loop frame invariant. A loop crossing several frames requires their transition matrices as well.

Theorem 21.17 (Parallel transport · *proved*)**.** *Parallel transport is the unique solution*

$$\dot{U}(t) = -a(t)U(t), \qquad U(0) = I.$$

Under a change of frame it becomes

$$U'(\gamma) = h(\gamma(1))^{-1}U(\gamma)h(\gamma(0)).$$

For the counter-clockwise coordinate square of side ϵ based at 0,

$$\mathrm{Hol}(\square_\epsilon) = I - \epsilon^2 F_{xy}(0) + O(\epsilon^3).$$

[2] Both conventions use $D = \mathrm{d} + A$ and transport $\exp(-\int A)$.

Proof. The preceding construction proves existence, uniqueness and invertibility. The endpoint formula follows by differentiating $h(\gamma(t))^{-1}U(t)h(\gamma(0))$.

For a scalar connection, Stokes gives

$$\mathrm{Hol}(\square_\epsilon) = \exp\left(-\int_{[0,\epsilon]^2} F\right) = 1 - \epsilon^2 F_{xy}(0) + O(\epsilon^3).$$

For matrices, first-order terms from opposite edges still cancel. Transverse changes give the derivative terms in F_{xy}. Products of first-order terms give its commutator. We retain their order in the four edge expansions.

For the square, write $C = A_x(0)$ and $D = A_y(0)$. Write $C_x = \partial_x A_x(0)$ and $D_y = \partial_y A_y(0)$. Here D denotes a constant matrix only within this calculation. Starting along positive x, integrate each edge equation through second order. The series bound makes all terms with at least three integrals $O(\epsilon^3)$:

$$\begin{aligned}
U_1 &= I - \epsilon C + \frac{\epsilon^2}{2}(C^2 - C_x) + O(\epsilon^3),\\
U_2 &= I - \epsilon D + \epsilon^2(D^2/2 - \partial_x A_y(0) - D_y/2) + O(\epsilon^3),\\
U_3 &= I + \epsilon C + \epsilon^2(C^2/2 + C_x/2 + \partial_y A_x(0)) + O(\epsilon^3),\\
U_4 &= I + \epsilon D + \frac{\epsilon^2}{2}(D^2 + D_y) + O(\epsilon^3).
\end{aligned}$$

For example, on the second edge $a(t) = D + \epsilon\partial_x A_y(0) + tD_y + O(\epsilon^2)$, for $0 \le t \le \epsilon$. The first integral contributes $-\epsilon D - \epsilon^2\partial_x A_y(0) - \epsilon^2 D_y/2$. The iterated integral contributes $\epsilon^2 D^2/2$. The other edges follow from their signed tangents in the same way. The successive transport acts on the left, giving

$$\begin{aligned}
U_4U_3U_2U_1 &= I + \epsilon^2(\partial_y A_x(0) - \partial_x A_y(0) - [C, D]) + O(\epsilon^3)\\
&= I - \epsilon^2 F_{xy}(0) + O(\epsilon^3).
\end{aligned}$$

■

The figure fixes the traversal order used in this calculation.

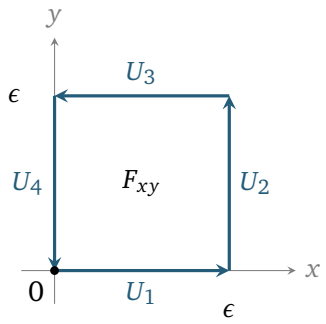


Figure 21.4: The path starts at 0 and traverses positive x, positive y, negative x, then negative y. Its transport is $U_4U_3U_2U_1$.

For the tangent connection, the square theorem measures the leading change $-\epsilon^2 R(\partial_x, \partial_y)$ of a transported vector. On a surface, choose positive coordinates orthonormal at the base point. The dictionary then gives $I + \epsilon^2 K(0)J + O(\epsilon^3)$, a rotation through the signed angle $K(0)\epsilon^2$ to leading order. This agrees with Lecture 18's curvature integral.

Recall that $\gamma \cdot \delta$ traverses γ first. Uniqueness therefore gives

$$U(\gamma \cdot \delta) = U(\delta)U(\gamma), \qquad U(\gamma^{-1}) = U(\gamma)^{-1}.$$

Transport reverses the order of the path product.

21.5 Flat connections and the Aharonov–Bohm phase

Definition 21.18. A connection is *flat* if $F = 0$.

This condition removes local loop curvature but need not remove holonomy around a hole.

Theorem 21.19 (Flat holonomy · *sketched; proved for U(1)*)**.** *For a flat connection, based-loop holonomy depends only on the based homotopy class. After choosing a base G-frame, the map*

$$[\gamma] \longmapsto \mathrm{Hol}(\gamma)^{-1}$$

is a homomorphism $\pi_1(M, x_0) \to G$, where G is the matrix structure group. For each based loop its conjugacy class is independent of the chosen base frame.

Proof for U(1). Write the based homotopy as $H(s,t)$, with t the path parameter. We subdivide its parameter square into rectangles on which transport depends only on endpoints. Each rectangle has unit boundary transport, so internal edges cancel. The extension to continuous paths makes this argument apply to continuous based homotopies. In a small ball with a unitary frame, $\mathrm{d}A = 0$. By Theorem 13.11, write $A = \mathrm{d}f$, with imaginary f. For endpoints p_0 and p_1 in this ball, transport is $\exp(f(p_0) - f(p_1))$. Equivalently, Stokes gives unit transport around every loop bounding a disk in that ball. The endpoint formula also defines local transport on continuous paths. Subdividing into these balls extends it to continuous paths, agreeing with ODE transport on piecewise smooth paths. Refinements do not change it, since endpoint factors cancel. The local frames $e_i e^{-f_i}$ have zero connection form. On a connected overlap they therefore differ by a constant phase, by the frame law. Either local formula gives the same fibre map.

The figure shows the parameter square, rather than a region embedded in M.

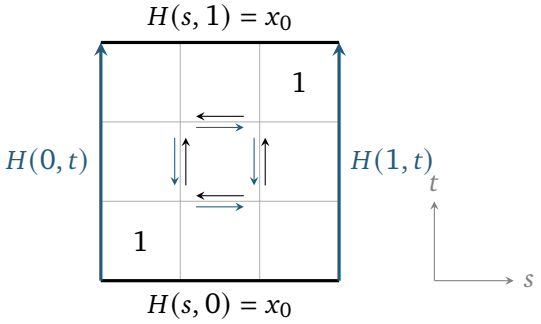


Figure 21.5: Each cell maps into one framed ball where $A = \mathrm{d}f$. After internal edges cancel in opposite pairs, only the two loop transports remain.

Given a continuous based homotopy, the Lebesgue lemma subdivides its parameter square into rectangles whose images lie in these balls. Choose a unit basis in the fibre at every grid vertex. Each rectangle has boundary product 1 by the local endpoint formula. Multiply the scalar boundary products. Every internal edge cancels with its reverse. The edges $t = 0$ and $t = 1$ are constant at the base point and have transport 1. The transports of the loops on $s = 0$ and $s = 1$ are therefore equal. Finally, taking inverses in the first-path-first composition rule gives the claimed homomorphism. This argument uses no global trivialization of the bundle. ∎

Lecture 5 assumed that a scalar statistical factor is homotopy invariant and multiplicative. By Theorem 21.19, inverse holonomy of a flat unitary line-bundle connection has both properties and is a unitary character of the fundamental group. Flatness remains a hypothesis on that statistical sector.

For a general matrix group, the variation of transport gives the corresponding sketch. Let $H(s,t)$ be a smooth based homotopy and $U(s,t)$ its transport from $t=0$ to t. In a pullback frame fixed along both endpoint edges, the based boundary conditions give

$$U(s,1)^{-1}\partial_s U(s,1) = -\int_0^1 U(s,t)^{-1}F(H_s,H_t)U(s,t)\mathrm{d}t.$$

To obtain it, differentiate $\partial_t U = -A(H_t)U$ in s and combine $\partial_s U + A(H_s)U$ before integrating in t. Patching frames and the passage to continuous homotopies are omitted here. If $F=0$, the formula makes the endpoint transport constant in s.

The angular one-form on the punctured plane is global although its angle is not:

$$\mathrm{d}\theta = \frac{-y\mathrm{d}x + x\mathrm{d}y}{x^2+y^2}, \qquad M = \mathbb{R}^2 \setminus \{0\}.$$

We model a long shielded solenoid by the removed point in this transverse plane. Its flux Φ is positive out of the page. The accessible exterior has zero electromagnetic field. A coherent source sends amplitudes through two slits, on opposite sides of the solenoid, to the same point on a screen. In this static model $\phi = 0$. The figure shows the two arms and a phase-resolved interference profile. The profiles assume equal unit arm magnitudes. Their apparatus phase difference is χ_{app}, and their flux phase is $\varphi_{\mathrm{AB}} = q\Phi/\hbar$.

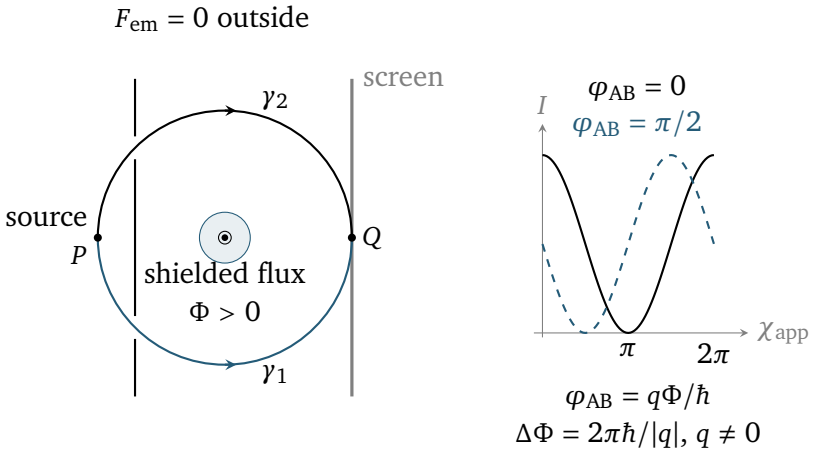


Figure 21.6: The field-free arms from source P to screen point Q form the counter-clockwise loop $\gamma_1 \cdot \gamma_2^{-1}$. The profiles plot $2+2\cos(\chi_{\mathrm{app}} + \varphi_{\mathrm{AB}})$ for $\varphi_{\mathrm{AB}} = 0$ and $\varphi_{\mathrm{AB}} = \pi/2$.

Example 21.20 (The Aharonov–Bohm connection)**.** Compute the holonomy on the trivial complex line over M with

$$A_{\mathrm{em}} = \frac{\Phi}{2\pi}\mathrm{d}\theta, \qquad A = -\frac{\mathrm{i}q\Phi}{2\pi\hbar}\mathrm{d}\theta.$$

Differentiation gives $\mathrm{d}(\mathrm{d}\theta) = 0$ on M, so $F = 0$ there. Let γ be the counter-clockwise unit circle, with $\theta(t) = 2\pi t$ and circulation 2π. Hence

$$\mathrm{Hol}(\gamma) = \exp\left(-\oint_\gamma A\right) = e^{\mathrm{i}q\Phi/\hbar}.$$

The loop does not bound a disk in M, so Stokes does not set this circulation to zero. The bundle is trivial, but the connection need not be gauge equivalent to the zero connection. Put $\alpha = q\Phi/(2\pi\hbar)$. On the angular cover $h = e^{i\alpha\theta}$ would cancel A. It descends to a single-valued map on M exactly when $\alpha \in \mathbb{Z}$. For an integer n, the genuine gauge map $h = e^{in\theta}$ changes α to $\alpha - n$ and leaves the holonomy unchanged. Thus these flat connections are distinguished by α modulo $\mathbb{Z}$.

The loop $\gamma_1 \cdot \gamma_2^{-1}$ winds once counter-clockwise, so its angular-form integral is also 2π by Lecture 3. Scalar transport gives $U(\gamma_1 \cdot \gamma_2^{-1}) = U(\gamma_1)U(\gamma_2)^{-1}$. By the coherent-arm physical input, the ratio of the two electromagnetic phases is $U(\gamma_1)U(\gamma_2)^{-1} = e^{iq\Phi/\hbar}$. For amplitudes c_1 and c_2 with the electromagnetic factors removed, the screen intensity is

$$\begin{aligned} I &= |c_1 U(\gamma_1) + c_2 U(\gamma_2)|^2 \\ &= |c_1|^2 + |c_2|^2 + 2\,\mathrm{Re}(c_1 \bar{c}_2 e^{iq\Phi/\hbar}). \end{aligned}$$

The phase is defined modulo 2π. The signed particle charge is q. For $q \neq 0$, the positive flux period is

$$\Delta\Phi = \frac{2\pi\hbar}{|q|}.$$

Changing flux changes the relative phase and hence shifts the interference fringes, although the field vanishes everywhere along both arms. The spatial displacement of the fringes also depends on the apparatus phases. The exterior curvature alone therefore does not determine the interference pattern. The holonomy retains the flux dependence. This is also the phase in *Quantum Mechanics II*, Lecture 3, for counter-clockwise forward hops on a ring.[3]

The phase lift in the figure displays the same holonomy on a cut angular interval.

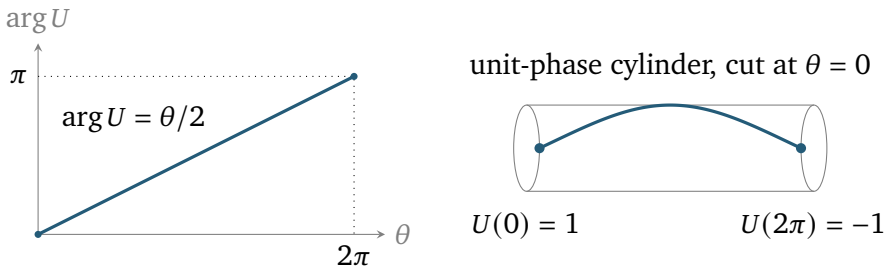


Figure 21.7: For $\alpha = 1/2$, transport along angle θ has phase $e^{i\theta/2}$. The two endpoints lie over the same base point but differ by the holonomy -1.

21.6 A constant nonabelian connection

The coefficient column σ is distinct from the Pauli matrices σ_x, σ_y and σ_z, recalled explicitly as

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For a unit column $v \in \mathbb{C}^2$, recall the Bloch vector with components $r_a = v^\dagger \sigma_a v$, for $a = x, y, z$. The action of SU(2) on these vectors is the rotation map of Lecture 4.

[3] Nakahara, §10.5.3, (10.95)–(10.100), uses $\hbar = 1$. Our potential convention is $A_{\mathrm{conn}} = -iqA_{\mathrm{em}}/\hbar$.

Example 21.21 (Curvature from noncommuting transport)**.** Find the curvature and small-square holonomy of the connection on $\mathbb{R}^2 \times \mathbb{C}^2$ given, for $a, b \in \mathbb{R}$, by

$$A = C\mathrm{d}x + D\mathrm{d}y, \qquad C = -\frac{\mathrm{i}a}{2}\sigma_z, \qquad D = -\frac{\mathrm{i}b}{2}\sigma_x.$$

Both coefficient matrices are traceless and anti-Hermitian, so their transport lies in SU(2). Indeed, unitarity follows as above and $\frac{\mathrm{d}}{\mathrm{d}t}\det U = -\operatorname{tr}(A(\dot\gamma))\det U = 0$. The exterior derivative is zero, but $[\sigma_z, \sigma_x] = 2\mathrm{i}\sigma_y$ gives

$$F = [C, D]\mathrm{d}x \wedge \mathrm{d}y = -\frac{\mathrm{i}ab}{2}\sigma_y \mathrm{d}x \wedge \mathrm{d}y.$$

It is nonzero when $ab \neq 0$. The constant edge equations have exact solutions, so

$$\mathrm{Hol}(\square_\epsilon) = e^{\epsilon D}e^{\epsilon C}e^{-\epsilon D}e^{-\epsilon C} = I + \frac{\mathrm{i}ab\epsilon^2}{2}\sigma_y + O(\epsilon^3).$$

The coefficient is $-F_{xy}$, as the transport theorem requires. Thus vanishing ordinary derivatives of a nonabelian potential do not imply flatness. Under Lecture 4's SU(2) $\to$ SO(3) map, the positive x edge rotates the Bloch vector by $-a\epsilon$ about z. The positive y edge rotates it by $-b\epsilon$ about x. The two return edges have the opposite angles in the same traversal order. Their product is a rotation about y with leading angle $-ab\epsilon^2$. Rotations about different axes need not commute.

The figure shows exact Bloch directions for one finite square of the connection computed in the example.

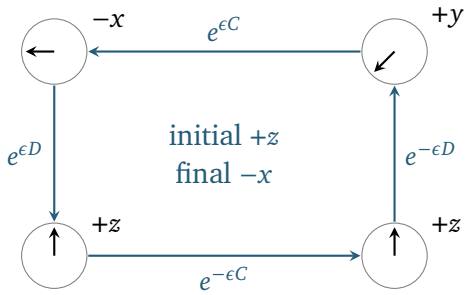


Figure 21.8: For $a = b = 1$ and $\epsilon = \pi/2$, the four transports send an initial Bloch direction $+z$ successively to $+z$, $+y$, $-x$ and $-x$. The final direction differs from the initial one despite the constant coefficients.

Self-check. Show that a pure gauge $A = h^{-1}\mathrm{d}h$ is flat and compute its transport.

Answer. The inverse derivative gives $\mathrm{d}(h^{-1}\mathrm{d}h) = -h^{-1}\mathrm{d}h \wedge h^{-1}\mathrm{d}h$, which cancels $A \wedge A$. This is the Maurer–Cartan identity for $\vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h$. Differentiating $U(t) = h(\gamma(t))^{-1}h(\gamma(0))$ verifies the transport equation and its initial condition. If h is global, every closed loop has holonomy I. The Aharonov–Bohm cancellation uses only a local h unless $\alpha \in \mathbb{Z}$.

Self-check. On two successive segments, each parameterized by a unit interval, let the coefficient matrices be

$$a_1 = -\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \qquad a_2 = -\begin{pmatrix} 0 & 0 \\ 3 & 0 \end{pmatrix}.$$

Compute the total transport. Then compute it in a new frame with endpoint values

$$h(0) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad h(1) = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

Here 0 and 1 label the endpoints of the complete path.

Answer. Each matrix squares to zero, so the edge transports are $I - a_1$ and $I - a_2$. Later transport multiplies on the left, giving

$$U = (I - a_2)(I - a_1) = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \qquad U' = h(1)^{-1} U h(0) = \begin{pmatrix} 1/2 & 3/2 \\ 3 & 10 \end{pmatrix}.$$

Reversing the factors instead gives $\begin{pmatrix} 7 & 2 \\ 3 & 1 \end{pmatrix}$, so the traversal order changes the answer.

Summary. A connection differentiates sections independently of their chosen frame. Its matrices change by $A' = h^{-1}Ah + h^{-1}\mathrm{d}h$. Curvature is $F = \mathrm{d}A + A \wedge A$, with $F' = h^{-1}Fh$ and $DF = 0$. Transport solves $\dot{U} = -A(\dot{\gamma})U$ and multiplies later segments on the left. A counter-clockwise small square has $\mathrm{Hol} = I - \epsilon^2 F_{xy} + O(\epsilon^3)$. A flat connection can retain nontrivial holonomy around a hole, as the Aharonov–Bohm phase $e^{\mathrm{i}q\Phi/\hbar}$ shows. Lecture 22 relates curvature flux to the transition winding of a line bundle.

Reading. Nakahara, §10.1.3–4, §10.3.4–5, §10.4.2 and §10.5.1, 10.5.3.